Multimedia Tools and Applications (2024) 83:39027–39049
https://doi.org/10.1007/s11042-023-16985-4

Robust image encryption algorithm in dwt domain

Rajwinder Kaur[1] · **Butta Singh**[1]

Received: 20 November 2022 / Revised: 22 August 2023 / Accepted: 11 September 2023 /
Published online: 7 October 2023
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2023

Abstract
Over the past few years, the use of chaotic maps have surfaced to become an important topic in the field of image encryption. The majority of chaos-based algorithms employed single or multiple rounds of confusion and diffusion. However, such algorithms increase the security and computational cost of the encryption process. On the other hand,a higher computational process is required to maintain security. To overcome this issue, a robust image encryption method using the features of two dimensional coupling map (2D-CM) and Discret Wavelet Transform(DWT) is proposed. Additionally, low-frequency component of DWT used for encryption can significantly save storage space thereby reducing computational overheads without compromising the security of the encryption. Finally, the cover image is combined with higher frequency coefficients to obtain the whole image, which will be totally secure against external attacks. The proposed encryption technique undergoes various security analyses, including Information Entropy (I_{en}), Correlation Coefficients (CC), Number of Pixel Change Rate (NPCR), Unified Average Change Intensity (UACI) and computational complexity and histogram analysis. The NPCR value and UACI obtained for all the images under simulation are greater than 99% and 33, respectively. Furthermore, results show that the proposed method has low correlation, high key sensitivity and sufficient key space (2^{256}) to withstand against external attacks.

Keywords DWT · Image encryption · Chaotic maps · NPCR · UACI

Abbreviations

I_{en}	Information Entropy
S	Cover Image
S''	Cipher Image
1D LM	One Dimensional Logistic map
2D-CM	Two dimensional coupling map
AES	Advanced Encryption Standard
CC	Correlation Analysis
CLS	Circular Left Shift

✉ Rajwinder Kaur
rajwinder086@gmail.com

[1] Department of Electronics and Communication Engineering, Guru Nanak Dev University Regional Campus, Jalandhar, India

CS	Compressive Sensing
DCT	Discrete Cosine Transform
DES	Data Encryption Standard
DWT	Discrete Wavelet Transform
FrFT	Fractional Fourier Transform
IDWT	Inverse Discrete Wavelet Transform
NPCR	Number of Pixel Change Rate
PWLCM	Piece Wise Linear Chaotic Map
SM	Sine Map
SVD	Singular Value Decomposition
UACI	Unified Average Change Intensity

1 Introduction

With the proliferation of networked technology, tremendous development has been recorded in videos, audio as well as images being media for illustrations of digital information to be communicated over the internet [1]. Among these media, images are of significant importance owing to their distinctive 2D data form, which includes personal information [2–4]. For example a medical image is a private image, as it includes information on health conditions. If this significant information gets detected in some unauthorized ways, extreme security issues may occur. Thus, such information in an image can be protected by image encryption by transforming it into an unrecognized form.

The simplest way of implementing image encryption is by understanding a digital image as a stream of binary, then after this data stream can be encrypted by employing a data encryption algorithm. These algorithms include Data Encryption Standard(DES) [5] and Advanced Encryption Standard (AES) [6]. It is worth mentioning here that image data exhibits many unique features like a large volume of data, strong redundancy and high correlation [7]. Considering an image as a binary stream will lead to the loss of these properties, thus making encryption ineffective. These issues have been resolved through several image encryption schemes proposed in the literature using different technologies [8]. Among these technologies, Chaotic map based image encryption methods are widely used in recent years due to fundamental features such as nonlinear, random and exactness [8]. Chaotic maps are used to generate secret keys, which can control, confuse and diffuse the encryption and decryption process. Implementing the appropriate chaotic map increases the complexity of the algorithm and can effectively resist attacks against confusion and diffusion. Some of the chaotic based image encryption techniques are discussed here. Baptista et al. [10] introduced a method of encryption based on logistics maps. In this method, the encrypted message is transferred to a text format consisting of different alphabets. These alphabets are mapped within the logistics map range, [0,1]. The main disadvantage is that it depends on the machine and the length of ciphertext is longer than plain text. Lian et al. [11] developed a confusion and diffusion based image encryption method to achieve better encryption results. The standard chaotic maps are exploited for shuffling and substitution processes. The standard map reduces the computational complexity and it can be implemented in the integer domain for image encryption. Wong et al. [12] introduced a better method than proposed by [11]. In this method, two rounds are done for both confusion and diffusion in order to obtain an effective image encryption algorithm. To improve the speed of the encryption process, another method of permutation and diffusion has been proposed by Patidar et al. [13]. In this work, the permutation and

diffusion are done in terms of rows and columns. Francois et al. [14] implemented a method based on a logistic map and integer values for the encryption process. The key space (2^{462}) is also increased as compared to the method proposed by [13] but the drawback of this method is that it has used 24 rounds for the encryption and decryption process. Zhou et al. [15] introduced an image encryption method using Piece Wise Linear Chaotic Map (PWLCM). The shuffling and substitution process is done on a bit level. The key length of this method is 2^{210} and it uses fewer rounds to encrypt the image. Pak et al. [16] proposed an improved Logistic and Sine map for image encryption. The encryption process is achieved by using the confusion and diffusion process. This method has extended the chaotic range of sine and logistic maps, but its key space is small as compared to the method proposed by Francois et al. [14]. Recently, Xiang et al. [17] (2020) employed a method based on improved one Dimensional Logistic map (1D LM) for image encryption. The results show that the proposed map has a wider dynamic range and high complexity. Also, the proposed method has high computational precision 2^{-16} which is sufficient to resist external attacks.

Image encryption methods based on the transform domain convert the pixel value of the cover image into the frequency domain, and then the positions of the coefficients are changed according to a specified algorithm. Then, by employing inverse transform one can obtain an encrypted image. Some of the transform domain encryption techniques are elucidated in this section. Zhu et al. [18] proposed Fractional Fourier Transform (FrFT) based image encryption algorithm. Here, the duration of FrFT is extended to any integer value. Hence, the transformation index and index of duration is considered as two secret keys. Liu et al. [19] presented a Discrete Cosine Transform (DCT) based encryption method. DCT was used to represent the pixel value in a real domain. This method has the advantage that the security of the encrypted image was enhanced. Kong et al. [20] developed a cascaded fractional Fourier transform based multi-image encryption method. In this technique, multiple secret keys are utilized to obtain the encrypted image. Because of multiple secret keys, this method can be used for multi-user authentication. Chen et al. [21] proposed an image encryption method based on Arnold transform and using Singular Value Decomposition (SVD). The cover image was converted using FrFT into the fractional domain. Then, SVD is applied to divide the fractional coefficients into three parts. The decrypted image was reconstructed by applying inverse Arnold transform and inverse FrFT on the encrypted image.

DWT is an important transform with the multiresolution feature of describing the spatial and frequency domain details and is used significantly in image processing. One can see in Fig. 1, the cover image (S) can be segregated into low frequency (LL), diagonal detail (HH), vertical detail (LH) and horizontal detail (HH) [22].

Most of the energy of the image (or the signal under observation) is concentrated in the low frequency region and a little amount of information is contained in the higher frequency coefficients. Thus, the energy contraction and multiresolution feature of DWT has motivated researchers to exploit it in image encryption algorithms. For example, Dang et al. [23] proposed a hybrid encryption scheme based on DWT and block cipher method. It employed DWT

Fig. 1 DWT coefficients

LH	HH
HH	HL

for image compression and block cipher for image encryption. Taneja et al. [24] employed a combined method based on chaotic stream cipher using fractional wavelet transform. Kong et al. [25] developed an image encryption technique formulated on the basis of FrFT and optical wavelet transform. Luo et al. [26] employed an encryption scheme based on chaotic map and wavelet transform. Nevertheless, these encryption schemes have insufficient key length for providing security to the image encryption scheme.

In image encryption, the conventional methods such as AES, DES are insufficient for protecting the digital images due to high redundancy. On the other hand, large computational time is required to complete the encryption process. Lian et al. [27] proposed encryption of the sign bit of the DC coefficients as well as all AC coefficients by utilizing a spatiotemporal chaos system. Xin et al. [28] proposed a logistic map and a technique based on the DCT domain to improve the security of [27] by incorporating a unique permutation in each DCT block. But, this permutation technique does not lead to improvement in the sensitivity of plain text in cipher images. Yu et al. [29] described a chaotic map and wavelets transform based image encryption. In this scheme, the wavelength function converts the image into a lower and higher frequency sub band. The low-frequency sub-band is considered to be an important sub-band for achieving better image quality for chaos encryption. The higher frequency coefficients of wavelet are XORed with encrypted low frequency coefficients. Then the encrypted areas are smoothen using Arnold Transform. The implemented method has a significant key space (2^{128})to resist attacks . Chaos map algorithm for single domain image encryption is recommended by Ramahrishnan et al. [30]. Frequency domains and time domains make the procedure more secure. The chaos map algorithm has better features and produces random images to encrypt the image more secure. The transformation unit uses the tent mapping technique and the substitution unit uses the Bernoulli map to follow DFT and DWT for the image. Wang et al. [31] introduced a hybrid form of coefficient transform multi-chaos encryption. The DWT is converted to produce metrics that are recombined to scramble. The scrambled image is encrypted using multiple chaos matrices. The initial condition and system parameters are sensitive to the sizeable key space and the algorithm calculation is minimal and it is secure against external attacks. Abdmouleh et al. [32] introduced a novel DWT-based selective medical image based encryption. This method combines encryption and compression to securely preserve and store medical images. This method is faster because it allows for significant gains in encryption- decryption processing time and encrypts only 6.25% coefficients of DWT. Zhu et al. [33] suggested a scheme of digital image compression as well as encryption which was based on cyclic shift and the theory of CS, using random Gauss matrix and sparse transformation for digital image compression, followed by the application of diffusion operation and cyclic shift on CS phase [33]. Lu et al. [34] proposed a hybrid technique based on hyperchaotic map and DNA encoding to encrypt the cover image. The security provided with this technique is very high but the speed of the DNA encoding has to be improved. Another algorithm based on hyper chaotic is employed by Kaur et al. [35] to create large secret key. The disadvantage of this method is that it pays less attention to computational complexity. From the above discussion, we have concluded that enhanced security of the encryption system comes with the increased computational cost of the system. Further, concluded from the discussion that the hybrid method yields more promising results as compared to the singular method. The contributions of the proposed method are as follows:

1. In this paper, we have proposed a hybrid method based on the features of chaotic map and DWT. Where 2D-CM helps in maintaining the security of the algorithm and the DWT maintains the cost of the system due to its energy compaction property.

2. 2D methods were implemented to overcome the drawbacks of a 1D chaotic system. These approaches outperform in terms of randomness and nonlinearity and are difficult to predict due to the large key space.
3. Integration of Circular Left Shift(CLS) with 2D-CM, removes the correlation between the neighbouring pixels and provide a high level of security to the encryption algorithm.
4. The computational time for the encryption and decryption process is also reduced.

2 Preliminaries

This section reviews DWT and 2D-CM, which are the basic components exploited in the proposed image encryption algorithm.

2.1 2D Coupling Map (CM)

Logistic Map(LM) and Sine Map(SM) are the two deterministic 1D non-linear maps exhibiting complex chaotic behaviour, utilized in various image encryption algorithms [36]. Mathematically, they are written in (1) and (2):

$$LM : X_{r+1} = \alpha X_r(1 - X_r) \tag{1}$$

$$SM : X_{r+1} = \beta Sin(\pi X_r) \tag{2}$$

Where (α,β) and (X_r and Y_r) are the control and initial parameters, respectively lies in the range [0,4]. But, the chaotic behaviour of both maps lies within (3.57, 4]. To enhance the chaotic range, 2D-CM is used in this proposed work and are expressed in (3) and (4) respectively.

$$CM(X_{r+1}) = \alpha(Sin((\pi Y_r) + \gamma)X_r(1 - X_r)) \tag{3}$$

$$CM(Y_{r+1}) = \alpha(Sin((\pi X_{r+1}) + \gamma)Y_r(1 - Y_r)) \tag{4}$$

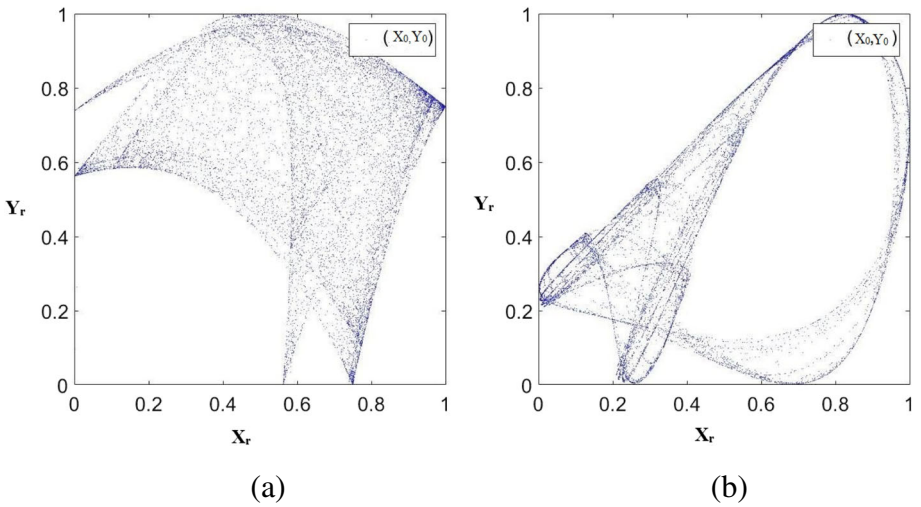


Fig. 2 Trajectories of (a) 2D-CM (b) 2D-LM

2D-CM is obtained using LM and SM. The output of LM is modulated by Sine and γ parameters to enhance its chaotic range. Here, the value of γ is set equal to 3. The performance of 2D-CM (3 and 4) and 2D-LM (5 and 6) is depicted in Fig. 2(a) and 2 (b) by plotting its trajectories, respectively.

$$LM(X_{r+1}) = \alpha(3Y_r + 1)X_r(1 - X_r) \tag{5}$$

$$LM(Y_{r+1}) = \alpha(3X_{r+1} + 1)Y_r(1 - Y_r) \tag{6}$$

In Fig. 2(a) and (b), it can be observed that the trajectory of 2D-CM distributes in a much wider region as compared to 2D LM. This means that 2D-CM can generate more random outputs with better erogicity properties.

3 Proposed method

Here in this section I discussed the block diagram (Fig. 3) and discussed in detail the explanation of DWT as well as 2D-CM based image encryption algorithm (DWCM-IE). First, S is transformed using DWT. Then, CM_1 and CM_2 chaotic matrices are generated using 2D-CM to perform two rounds of confusion and the diffusion process using (3) and (4). The

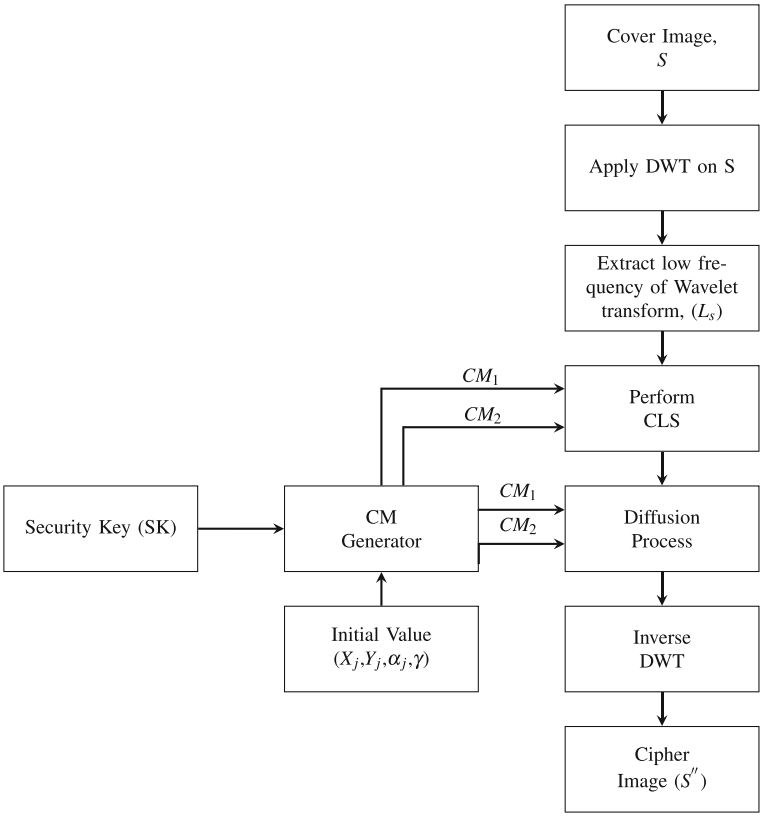


Fig. 3 The Block diagram of a chaotic map and DWT based image encryption process

2D-CM confusion can efficiently break the correlation among neighbouring pixels, also pixel values are changed completely through tdiffusion process. After that, inverse DWT (IDWT) is applied to convert S into the whole cipher image S''.

3.1 DWT of *S*

In this proposed algorithm, S is decomposed into four parts: Approximation Coefficients (L_s), Horizontal Coefficients (H_h), Vertical Coefficients (V_h), and Diagonal Coefficients (H_l), using Daubechies (db1) Ist level decomposition of wavelet transform. After decomposition, we have extracted L_s coefficients which are deemed as S'' for encryption.

Algorithm 1 Encryption algorithm (DWCM-IE).

Input: Secret key $SK = (x_0, y_0, \alpha, D, H_1, H_2)$, Cover image, S, Discrete Wavelet coefficients L_s
Output: Cipher Image, S''
Step 1. Apply DWT on S to obtain the frequency coefficients.
L_s, H_h, V_h, H_l = DWT(S)
Step 2. Generate two intial values X_{01}, Y_{01}, α_1 and X_{02}, Y_{02}, α_2 using Algorithm 2.
Step 3. CM_1 and CM_2 are generated using two initial conditions ((3) and (4)).
Step 4. Apply CLS to the L_s coefficients using CM_1 and CM_2
Step 5. Do substitution of rows and columns using (7)
Step 6. Concatenate the lower and higher wavelet coefficients.
Step 7. Apply IDWT to generate S''.
Ouput: S''.

3.2 Secret key generation

To generate the initial states of the 2D-CM, a secret key (SK) is used. The chaotic matrices generated by 2D-CM are then utilized to execute the permutation and diffusion process. The chaotic map based encryption algorithm should have a secret key length greater than 100 bits in order to resist brute attacks [37]. In thr proposed algorithm, the length of the secret key is set to 256 bits, $SK = (x_0 = 0.0230, y_0 = 0.3251, \alpha = 0.8059, H = 0.8565, D_1 = 6367817, D_2 = 7802400)$ according to the rapid development of computing ability of the computer. x_0, y_0, H, D_1, D_2 and α are the initial as well as control parameters, respectively. The key space of the proposed method is enlarged and the initial value is changed using the value of H, D_1 and D_2. The x_0, y_0, H and α are decimal numbers having 52 bits using the IEEE 754 Floating-165 Point standard. The pseudocode for the generation of initial parameters is as in Algorithm 2.

3.3 Circular left shift (CLS)

A digital image has high redundancy and high correlation between the neighbouring pixels, as the digital images are represented in 8 bits. The proposed method utilizes CLS to disintegrate correlation among neighbouring pixels. In this method, the position of row (N) and column (M) is shuffled simultaneously in one operation. The detailed description is as follows, first S of size ($N \times M$) is chosen and on it, DWT is applied to extract L_s. Then a chaotic sequence

Algorithm 2 Generation of initial values for 2D-CM.

Input: Secret key(SK) of length 256 bits
Output: (X_1, Y_1, α_1) **and** (X_2, Y_2, α_2)
x_o=$(\sum_{j=1}^{52}(SK[j].2^{52-j})).2^{-52}$
y_0=$(\sum_{j=53}^{104}(SK[j].2^{52-j})).2^{-52}$
α=$(\sum_{j=105}^{156}(SK[j].2^{52-j})).2^{-52}$
H=$(\sum_{j=157}^{208}(SK[j].2^{52-j})).2^{-52}$
D_1=$\sum_{j=209}^{232}(SK[j]*2^{j-209})$
D_2=$\sum_{j=232}^{256} SK[j]*2^{j-232}$
while *j=1 to 2* **do**
  X_j=$mod(x_0 + D_j H, 1)$
  Y_j=$mod(y_0 + D_j H, 1)$
  α_j=$0.9 + (\alpha + D_j H, 0.1)$
end

(CM_1) is generated using (3) and (4) followed by sorting the values of CM_1 column-wise in ascending order $CM_1^{'}$. Then, index value I is obtained indicating the permuted locations of $CM_1^{'}$. Afterwards, the coefficients of L_s are selected using $I_{(1,1)}$, $I_{(1,2)}$, $I_{(1,3)}$,..., $(I_{(N,M)})$ into circular form. Then, the positions of L_s are left shifted. Thus, the shuffled circular shifted matrix $L_s^{'}$ is obtained.

Example of CLS Fig. 4 shows an example of CLS. The shuffling process by CLS is explained as follows:

1. The first row of I is (2,1), (3,2), (2,3) and (2,4). Select the coefficients L_s in red, namely (2, 7, 10, 14) and shift them to one position left. The results obtained after shifting process are $L^{'}_{s(2,4)} = L_{s(2,1)}$), $L^{'}_{s(2,1)} = L_{s(3,2)}$, $L^{'}_{s(2,3)} = L_{s(2,4)}$ as shown in Fig. 4(e) in red .
2. Consider the second row of I as (3,1) (4,2), (3,3), (1,4), again select L_s in blue as (3, 8, 11, 13), and shift them two positions to the left. The result obtained after shifting process are $L^{'}_{s(3,3)} = L_{s(3,1)}$), $L^{'}_{s(1,4)} = L_{s(4,2)}$, $L^{'}_{s(3,1)} = L_{s(3,3)}$),$L^{'}_{s(4,2)} = L_{s(1,4)}$
3. The third row of I is (4,1), (1,2), (1,3), (4,4). again select L_s in green as (4, 5, 9, 16), and shift them three positions to the left. The result obtained after shifting process are $L^{'}_{s(1,2)} = L_{s(4,1)}$, $L^{'}_{s(1,3)} = L_{s(1,2)}$, $L^{'}_{s(4,4)} = L_{s(1,3)}$, $L^{'}_{s(4,1)} = L_{s(4,4)}$
4. The fourth row of I is (1,1), (2,2), (4,3), (3,4), again select L_s in white as (1, 6, 12, 15), and shift them four positions to the left. The result obtained after shifting process are $L^{'}_{s(1,1)} = L_{s(1,1)}$, $L^{'}_{s(2,2)} = L_{s(2,2)}$, $L^{'}_{s(4,3)} = L_{s(4,3)}$, $L^{'}_{s(3,4)}) = L_{s(3,4)}$. After the permuted operation, the resultant circular shifted values $(L_s^{'})$ is shown in Fig. 4(e).

3.4 2D-CM Diffusion

The process of encrypting two S (having minor differences) using the same security key can cause entirely different results in S'' is known as the diffusion process. A good diffusion property signifies that the encryption algorithm is efficient to resist external attacks from the eavesdropper. In this proposed encryption algorithm, we designed 2D-CM diffusion

0.9221	0.8342	0.8566	0.4842
0.4411	0.9886	0.6767	0.3424
0.5611	0.1613	0.7622	0.7732
0.7911	0.6198	0.9292	0.5252

(a)

0.4411	0.1613	0.6767	0.3424
0.5611	0.6198	0.7622	0.4842
0.7911	0.8342	0.8566	0.5252
0.9221	0.9886	0.9292	0.7732

(b)

2,1	3,2	2,3	2,4
3,1	4,2	3,3	1,4
4,1	1,2	1,3	4,4
1,1	2,2	4,3	3,4

(c)

1	5	9	13
2	6	10	14
3	7	11	15
4	8	12	16

(d)

1	4	5	8
7	6	14	2
11	10	3	15
16	13	12	9

(e)

Fig. 4 An example of shuffling lower frequency coefficients of image (S) using CLS process (a) Chaotic values generated by 2D-CM of size 4×4, CM (b) Sorted chaotic matrix, $CM^{'}$ (c) Index of the sorted matrix, I (d) Low frequency coefficients of image (S), L_s (e) Circular shifted matrix, $L_s^{'}$

in two modes: (a) row substitution and (b) column substitution. The diffusion process is mathematically defined as:

$$L_{s(j)}^{''} = \begin{cases} L_{s(j)}^{'} + L_{s(R)}^{'} + (CM_{(j)} * 2^{32}) modf & j = 1 \\ L_{s(j)}^{'} + L_{s(j-1)}^{''} + (CM_{(j)} * 2^{32}) modf & j\varepsilon[2, R] \end{cases} \tag{7}$$

Where f is the allowed frequency value in the DWT coefficients after converting its value into an unsigned integer value. $L_{(s)}^{'}$ are the shuffled coefficients. CM is the chaotic matrix of size $N \times M$. The maximum allowed value for f is 256. The value of $R = N$ and $R = M$ for row substitution and column substitution.

3.5 Inverse discrete wavelet transform (IDWT)

In the proposed encryption method, after CLS and 2D-CM diffusion process applied on the low frequency coefficients as ($L_s^{''}$). The $L_s^{''}$ coefficients are combined with the higher frequency coefficients and IDWT is performed to obtain the cipher image. $S^{''}$.

$$S^{''} = IDWT[L_s^{''}, H_h, V_h, H_l] \tag{8}$$

3.6 Decryption process

The decryption process on S'' is similar to the encryption process and is shown in Fig. 5.

The steps included in the decryption process are as follows:

1. The secret key SK is generated using Algorithm 2.
2. $S^{''}$ is converted into frequency domain using DWT.

$$[L_s, H_h, V_h, H_l] = DWT(S^{''}) \tag{9}$$

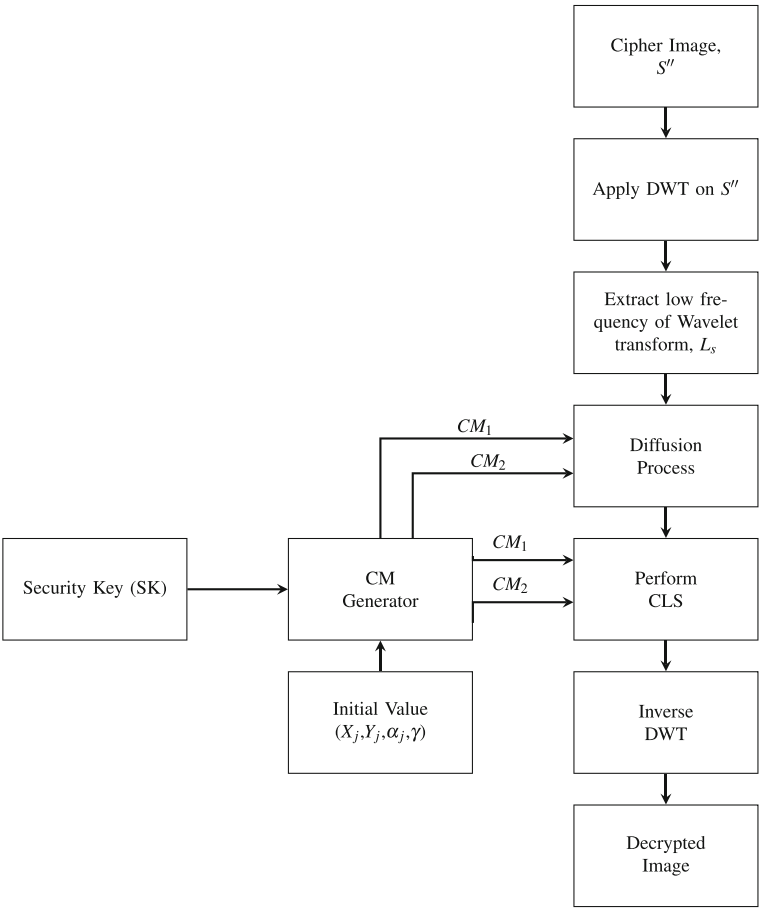


Fig. 5 Block diagram of a chaotic map and DWT based image decryption process

3. Extract the low frequency coefficients L_s and apply the diffusion process as

$$D_j^{''} = \begin{cases} L_{s(j)} - L_{s(R)} - (CM_j * 2^{32}) modf & j = 1 \\ L_{s(j)} - D_{j-1}^{''} - (CM_j * 2^{32}) modf & j \varepsilon [2, R] \end{cases} \tag{10}$$

4. Apply CLS on $D^{''}$ to shuffle the positions of L_s as

$$D^{'''} = CLS(D^{''}) \tag{11}$$

5. Apply IDWT after combining the lower as well as higher frequency coefficients to get S.

$$S = IDWT[D^{'''}, H_h, V_h, H_l] \tag{12}$$

Table 1 NPCR and UACI of different images

Image(512*512)	NPCR	UACI
Lena	99.9805	33.4349
Baboon	99.9832	33.5373
Airplane	99.9763	33.4866
Bridge	99.9714	33.6696
Peppers	99.9832	33.5946
Tank	99.9805	33.4245
Women	99.9832	33.5946
Cameraman	99.9863	33.6943
Couple	99.9840	33.6049
Boat	99.9790	33.5547
Elaine	99.9729	33.4481

4 Result and discussion

4.1 Resistance against differential attacks

It is an effective method, employed to break encrypted images. A robust diffusion property in an image encryption approach can withstand differential attack. Two parameters: (a) UACI and NPCR are deemed to evaluate the diffusion properties of an image encryption scheme [36]. In the proposed algorithm, Lena, Baboon, Airplane, Bridge, Pepper, Tank, Women, Cameraman, Couple, Boat and Elaine are the test images used to verify the performance of the proposed algorithm (Table 1).

For each test image, we change one bit of a randomly selected pixel to generate a new plaintext image, and then encrypt both plaintext images using the proposed encryption algorithms with the same security. The NPCR and UACI values obtained by all the test images are greater than 99 and 33, showing that the proposed encryption algorithm is very sensitive with respect to small changes in the plaintext, respectively.

4.2 Security analysis

An image encryption algorithm should have sufficient key space and key sensitivity to resist external attacks. A high key sensitivity indicates that a minor change in its parameter results in different encryption and decryption results.

The proposed DWCM-IE scheme has sufficient key space of 2^{256} with a key length of 256. This is sufficient to resist eavesdropper attacks. To verify the key sensitivity analysis, $SK_1 = SK(200) - 1$ and $SK_2 = SK(240) - 1$ are derived by changing one bit in SK. Here $SK(200)$ and $SK(240)$ represent the positions of the bit in SK. The cover image (Fig. 6(a)) is encrypted using SK and SK_1, the results of encryption shown in Fig. 6(b) and 6 (c) are totally different. Figure 6(d) shows the results of their difference. A decrypted image can be extracted using the correct key from S'' (Fig. 6(b)). The other two cipher images Fig. 6(c) and (d) are decrypted using one bit difference in the SK and the results are depicted in Fig. 6(f) as well as Fig. 6(g). Their different results are shown in Fig. 6(h). Thus, the proposed method is found to be highly sensitive to its security keys for encryption as well as decryption procedure.

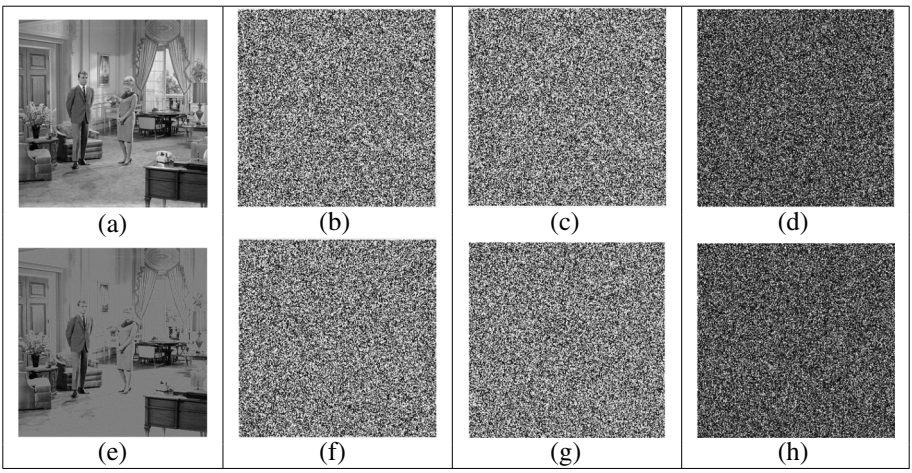


Fig. 6 Secret Key sensitivity analysis of encryption and decryption process (a) Cover Image, S; (b) Encrypted image using key SK (c) Encrypted image using key SK_2 (d) Difference of encrypted image:$|b - c|$ (e) Decrypted image using key SK (f) Decrypted image 'b' using key SK_1 (g) Decrypted image 'b' using key SK_2 and (h) Difference of decrypted images:$|e - f|$

4.3 Information entropy (I_{en})

How robust an encryption method is can be analyzed by I_{en}. The results of I_{en} are calculated using the proposed method using different images are listed in Table 2. The I_{en} value obtained after the encryption process for all the test images is nearer to the original I_{en} value of the cover images. But the I_{en} is decreased for all the images after the decryption process.

Table 2 Evaluation of entropy for different images

Images	Cover image	Encrypted	Decrypted
Lena	7.5983	7.0644	6.6923
Baboon	7.2925	7.0774	6.3720
Airplane	6.6776	7.0714	4.7966
Bridge	5.7056	7.0928	5.8810
Peppers	7.5839	7.0673	6.2239
Tank	6.5380	7.0551	6.2772
Women	7.2767	7.0343	6.0257
Cameraman	7.0097	7.1017	5.6413
Couple	7.2952	7.0756	6.3624
Boat	7.1914	7.0746	6.0036
Elaine	7.5107	7.0556	6.3624

4.4 Histogram analysis

Figures 7 and 8 show histogram analysis of S and S'' extracted using the proposed algorithm. The histogram of S contains large rises and declines, whereas the histogram of S'' for all the images is fairly uniform and totally different from histograms of their respective S. Hence, the proposed algorithm can withstand external attacks and an eavesdropper cannot detect the information through the histogram analysis.

4.5 Correlation analysis (CC)

Digital images exhibits some intrinsic properties like high correlation among neighbouring pixels and high redundancy. These can be utilized by eavesdropper to extract secret information. Thus correlation among neighbouring pixels is an essential parameter which needs to be determined.

In the proposed work, random 3000 adjacent pixels are selected from S and encrypted image to evaluate the horizontal, diagonal as well as vertical correlation coefficients. These correlations are calculated by the proposed algorithm and is tabulated in Table 3. Figures 9, 10, 11 and 12 demonstrates the correlation between cover (Boat, Jetplane, Tank, Lena, Pepper, Bridge, Couple, Woman, Cameraman, Baboon and Elaine) and their encrypted images. As one can conclude from the graphs that S exhibits high correlation which decreases after the encryption.

4.6 Computational complexity

The computational complexity of an algorithm is used to compute the time taken by the algorithm to complete the encryption/decryption process. The time analysis has been done on MATLAB R2017a running on Processor M2 pro, 10 core CPU, 16-Core GPU with a 16GB RAM computer. The encryption/decryption time taken by the proposed algorithm for images Lena, Baboon, Airplane, Bridge, Tank, Women, Cameraman, Couple, Boat and Elaine are shown in Table 4.

5 Comparison with existing techniques

Comparisons of the proposed encryption technique with other existing methods are listed in Tables 5 and 6, including NPCR, UACI and key length, respectively. One can see in Table 5, that the proposed technique exhibits higher NPCR as compared to [38, 39, 41–46] for all the images used for simulation. For image Cameraman, Lena, Baboon, Boat and Pepper, the proposed method shows higher value of UACI as compared to [39, 42].

But, the proposed methods lag behind the UACI (33.43, 33.44) as compared to [41] in the case of image Lena and Elaine, respectively. In comparison with the method employed by [38], the proposed method has a higher value of UACI (33.43, 33.69, 33.53, 33.55) for the images Lena, Cameraman, Baboon and Boat, respectively.

When speaking of key length, it should be greater than 2^{100} to withstand the brute attack [47]. In the proposed work, the key length is larger than [48–51] and equal to the method proposed by [52].

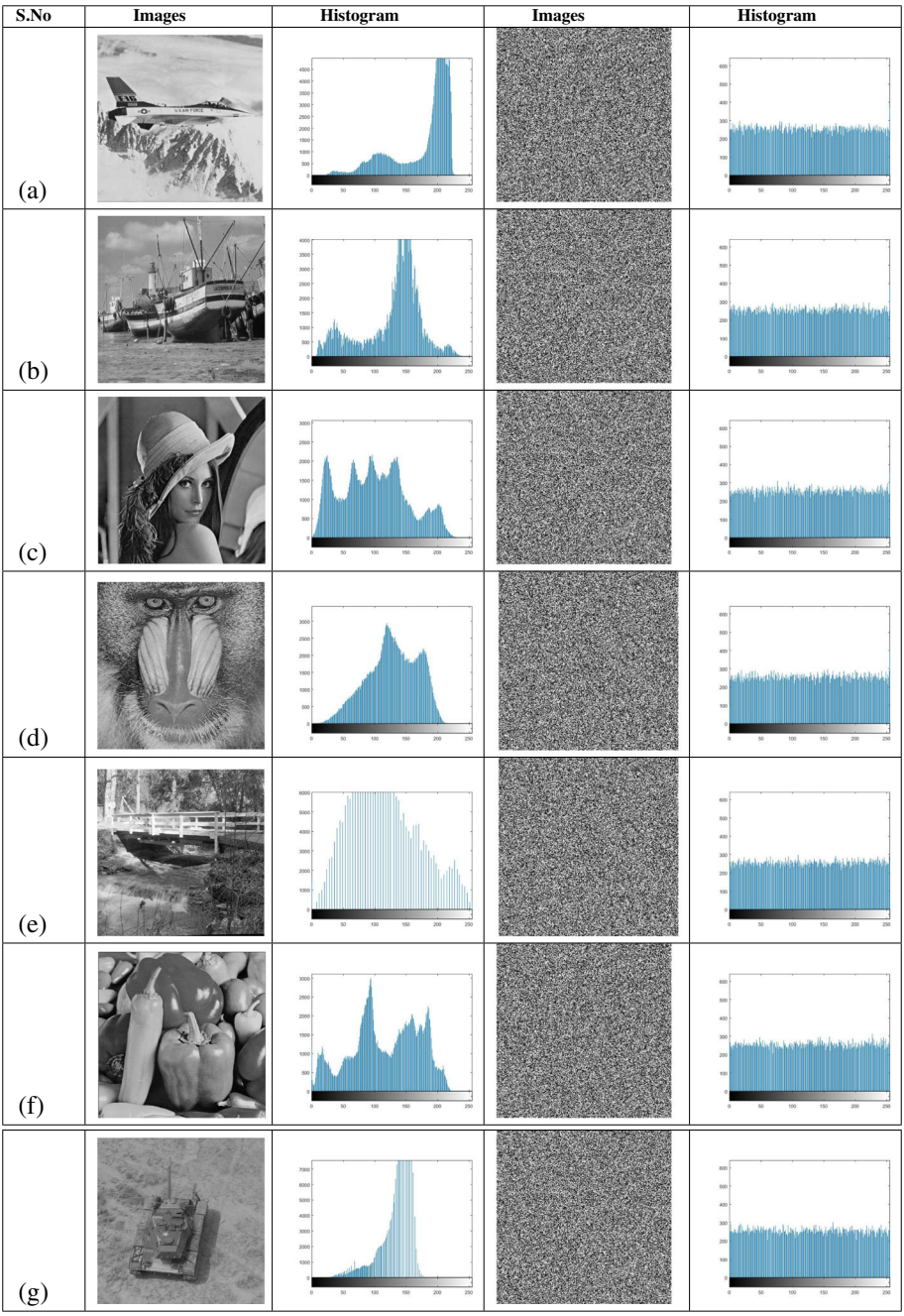


Fig. 7 Histogram analysis of cover (S) and its encrypted images (S'') (a) Jetplane (b) Boat (c) Lena (d)Baboon (e) Bridge (f) Pepper and (g) Tank

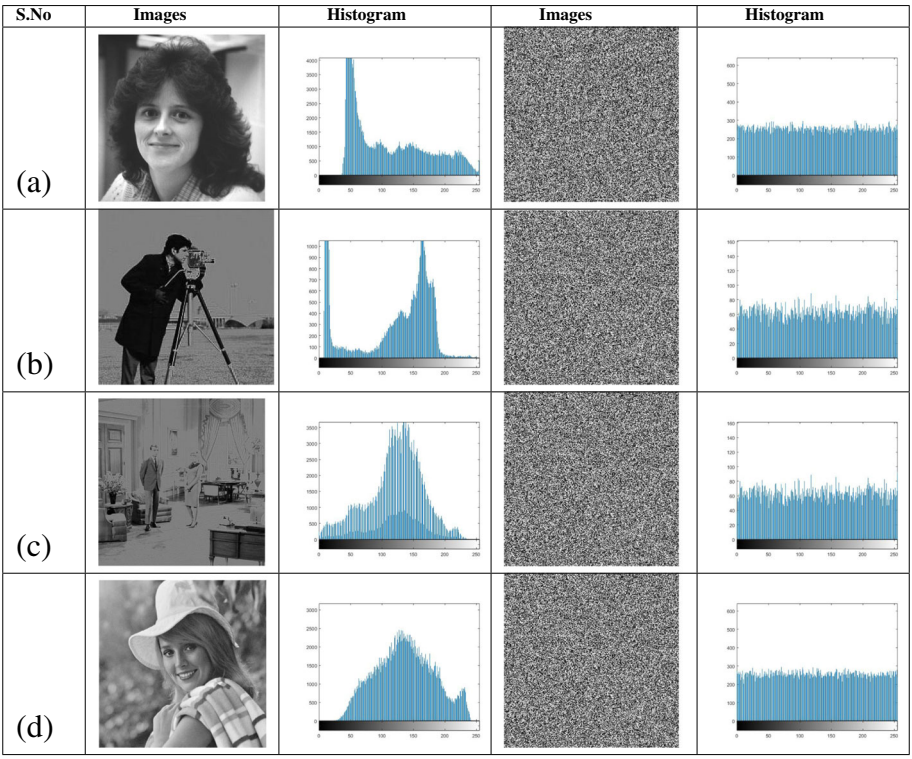


Fig. 8 Histogram analysis of cover (S) and its encrypted images (S'') (a) Woman (b) Cameraman (c) Couple (d)Elaine

Table 3 Correlation coefficients of different cover and encrypted images

Images	Cover			Encrypted		
Correlation	Horizontal	Vertical	Diagonal	Horizontal	Vertical	Diagonal
Couple	0.9409	0.9521	0.9190	-0.0048	-0.0365	0.0015
Cameraman	0.9354	0.9607	0.9066	-0.0197	-0.0055	-0.0145
Lena	0.9695	0.9852	0.9500	-0.0192	0.0017	0.0094
Elaine	0.9791	0.9752	0.9774	-0.0252	0.0346	0.0021
Tank	0.8757	0.8832	0.8341	-0.0081	-0.0229	0.0059
Women	0.9968	0.9965	0.9944	-0.0259	0.0129	0.0165
Airplane	0.9695	0.9649	0.9372	0.0140	-0.0108	0.0035
Boat	0.9330	0.9704	0.9131	0.0094	0.0151	-0.0549
Pepper	0.9725	0.9734	0.9440	-0.0155	-0.0075	-0.0568
Bridge	0.9361	0.4442	0.1910	-0.0009	-0.0180	0.0013
Baboon	0.9270	0.9062	0.8604	-0.0077	-0.0048	0.0151

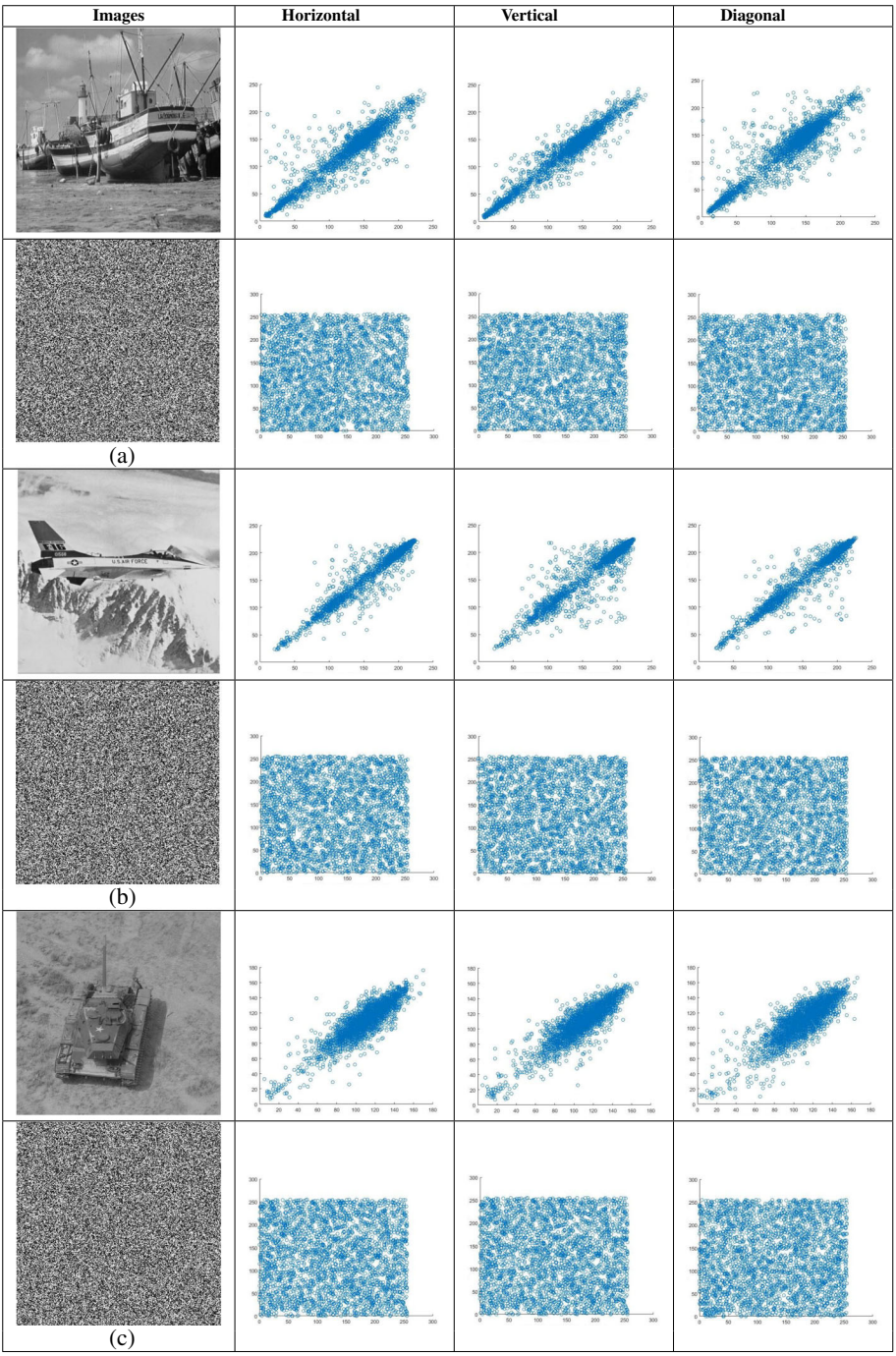


Fig. 9 Horizontal, vertical and diagonal correlation of cover(S) and its encrypted images (S'') for (a) Boat (b) Jetplane and (c) Tank

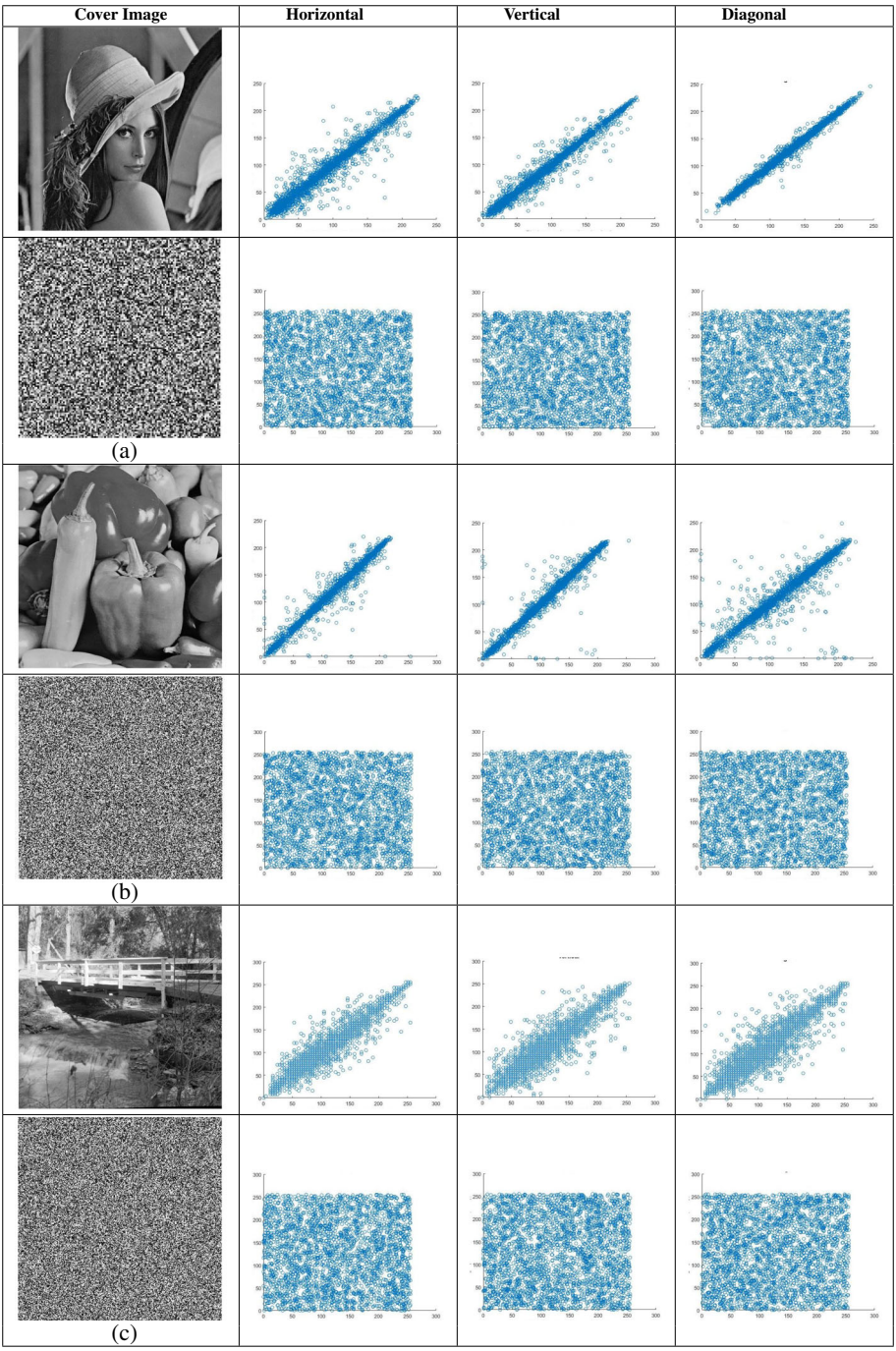


Fig. 10 Horizontal, vertical and diagonal correlation of cover (S) and its encrypted images (S'') for (a) Lena (b) Pepper and (c) Bridge

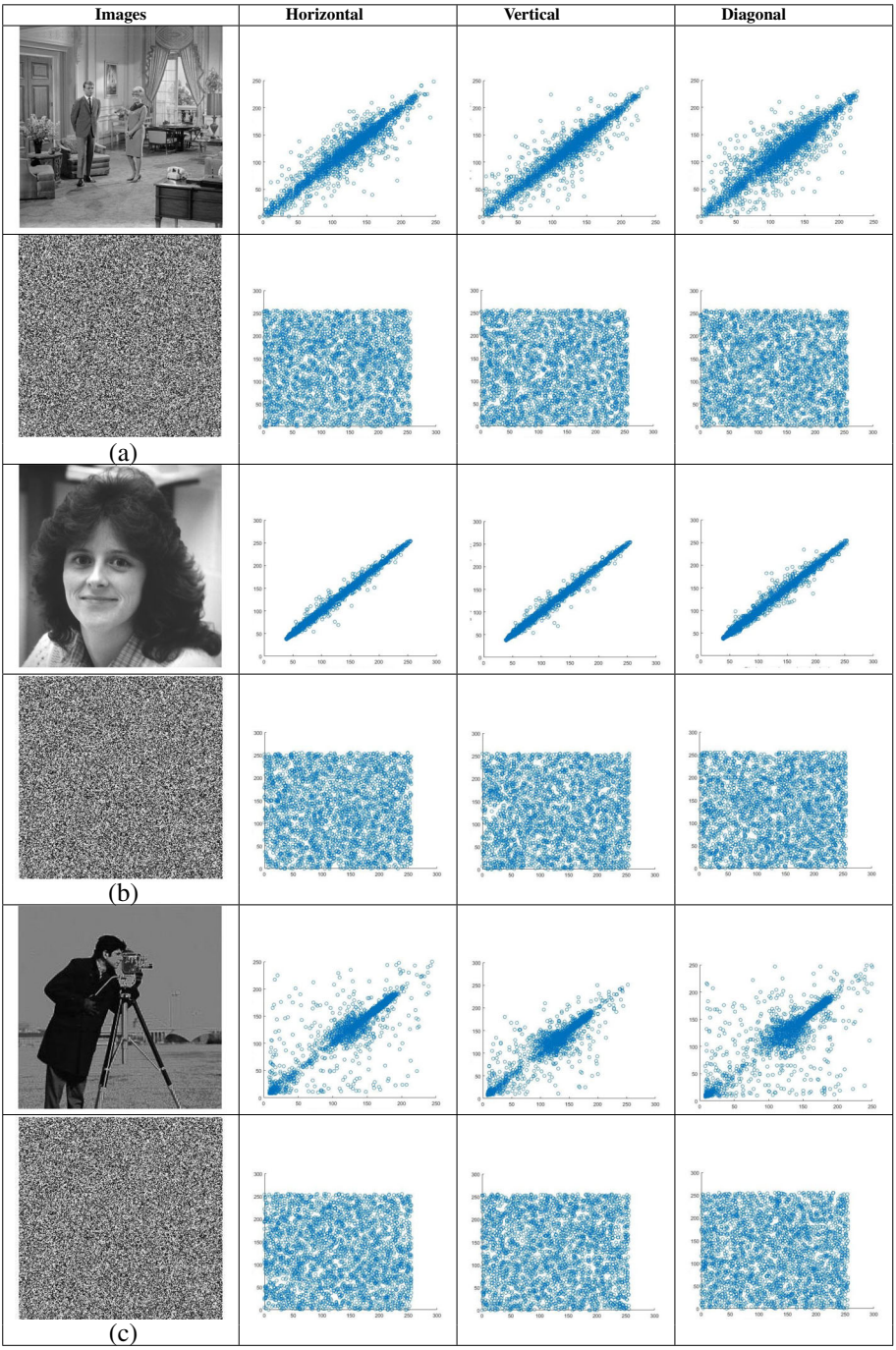


Fig. 11 Horizontal, vertical and diagonal correlation of cover (S) and its encrypted images (S'') for (a) Couple (b) Woman and (c) Cameraman

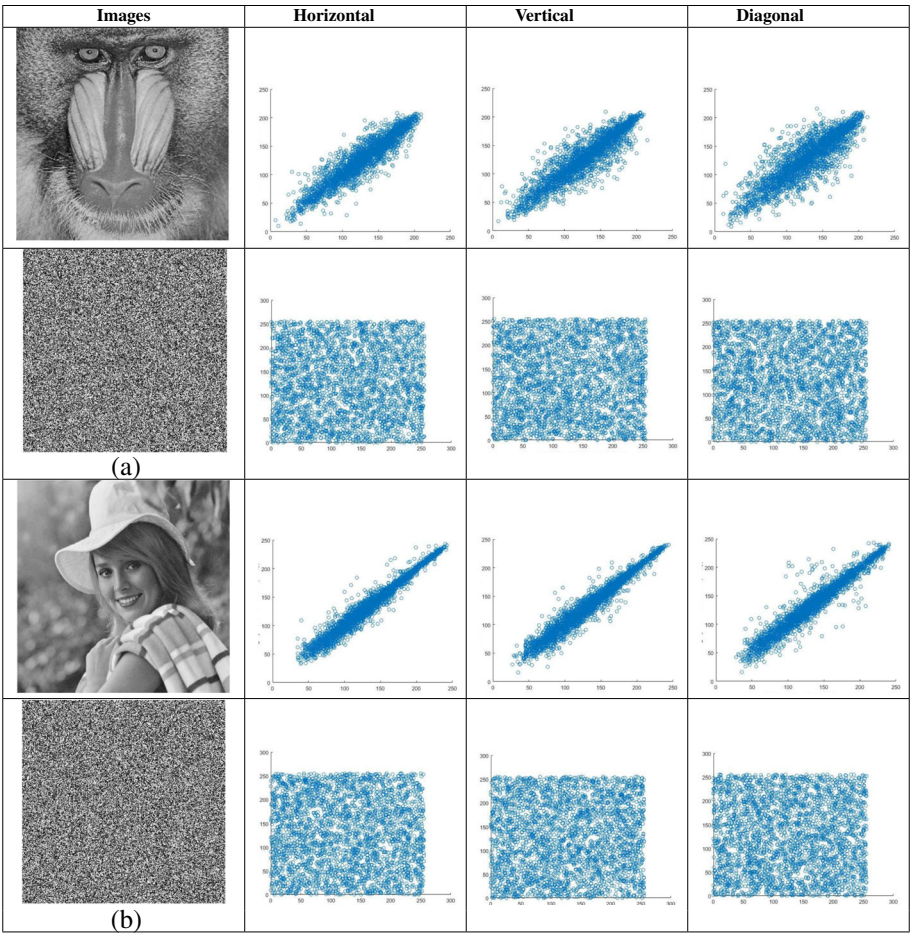


Fig. 12 Horizontal, vertical and diagonal correlation of cover (S) and its encrypted images (S'') for (a) Baboon (b) Elaine

Table 4 Time taken (seconds) by the proposed method for the encryption and decryption process

Images	Encryption time	Decryption time
Lena	0.7905	0.6105
Baboon	0.9215	0.7195
Airplane	0.8295	0.5995
Bridge	0.9215	0.7055
Tank	0.7245	0.6385
Women	0.7575	0.7065
Cameraman	0.6830	0.8145
Couple	0.6765	0.6805
Boat	0.7845	0.5705
Elaine	0.7575	0.5665

Table 5 Comparative analysis of NPCR and UACI with existing algorithms

Images	Authors	NPCR	UACI
Lena	Brindha et al. [38]	99.59	33.07
	Wang et al. [39]	99.60	33.41
	Norouzi et al. [41]	99.66	33.55
	Zarebnia et al. [42]	99.63	33.42
	Luo et al. [43]	99.63	33.69
	Zhu et al. [46]	99.63	33.60
	Proposed	99.98	33.43
Cameraman	Brindha et al. [38]	99.60	33.22
	Wang et al. [39]	99.61	33.40
	Norouzi et al. [41]	99.64	33.40
	Proposed	99.98	33.69
Baboon	Brindha et al. [38]	99.27	32.17
	Wang et al. [39]	99.55	33.47
	Norouzi et al. [41]	99.63	33.63
	Luo et al. [41]	99.61	33.43
	Hosny et al. [44]	99.60	33.47
	Hua et al. [45]	99.59	33.52
	Zhu et al. [46]	99.67	33.42
	Proposed	99.98	33.53
Boat	Brindha et al. [38]	99.59	32.92
	Wang et al. [39]	99.58	33.48
	Alawida et al. [40]	99.75	33.37
	Hosny et al. [44]	99.61	33.46
	Hua et al. [45]	99.60	33.47
	Zhu et al. [46]	99.64	33.49
	Proposed	99.97	33.55
Pepper	Zarebnia et al. [41]	99.66	33.57
	Luo et al. [43]	99.62	33.43
	Hosny et al. [44]	99.58	33.40
	Hua et al. [45]	99.61	33.55
	Zhu et al. [46]	99.61	33.51
	Proposed	99.98	33.59
Elaine	Zarebnia et al. [41]	99.65	33.47
	Alawida et al. [40]	99.74	33.37
	Luo et al. [43]	99.60	33.53
	Proposed	99.97	33.44

Table 6 Comparative analysis of key length with existing algorithms

Authors	Basu et al. [48]	Bansal et al. [49]	Murillo et al. [50]	Barker et al. [51]	Mousa et al. [52]	Proposed
Key length	2^{128}	2^{128}	2^{128}	2^{168}	2^{256}	2^{256}

6 Conclusions

In this work, a robust image encryption method using the features of 2D-CM and DWT is proposed. First, we converted the cover image into frequency coefficients using DWT and then permutation is applied on the L_s coefficients to break the correlation between the neighbouring frequency coefficients. The obtained permuted coefficients were then diffused to change the value of the L_s coefficients. Further, to enhance the security, twice permutation and confusion are implemented. Sensitivity analysis shows that the single bit change in secret key results in different decrypted image. In future, encryption based on filtering techniques combined with features of chaos and transform domain can be utilized to improve the image encryption and decryption process and provide protection from intruders.

Availability of data and material The data analysed during the current study is available at USC-SIPI database maintained by university of southern california (https://sipi.usc.edu/database/database).

Declarations

Conflict of interest The authors declare no conflict of interest, financial or otherwise.

References

1. Kaur M, Kumar V (2020) A comprehensive review on image encryption techniques. Arch Comput Methods Eng 27(1):15–43
2. Lu Q, Zhu C, Deng X (2020) An efficient image encryption scheme based on the LSS chaotic map and single S-box. IEEE Access 8:25664–25678
3. Nazari K, Ebadi MJ, Berahmand K (2022) Diagnosis of alternaria disease and leafminer pest on tomato leaves using image processing techniques. J Sci Food Agric 102(15):6907–6920
4. Rostami M, Berahmand K, Nasiri E, Forouzandeh S (2021) Review of swarm intelligence-based feature selection methods. Eng Appl Artif Intell 100:104210
5. FIPS PUB 46, (1999) Data encryption standard (DES)
6. FIPS PUB 197, (2001) Advanced encryption standard (AES)
7. Yu SS, Zhou NR, Gong LH, Nie Z (2020) Optical image encryption algorithm based on phase-truncated short-time fractional Fourier transform and hyper-chaotic system. Opt Lasers Eng 124:105816
8. Ghadirli HM, Nodehi A, Enayatifar R (2019) An overview of encryption algorithms in color images. Signal Process 164:163–85
9. Dhall S, Pal SK, Sharma K (2018) Cryptanalysis of image encryption scheme based on a new 1D chaotic system. Signal Process 146:22–32
10. Baptista M (1998) Cryptography with chaos. Phys Lett 240(1–2):50–4
11. Lian S, Sun J, Wang Z (2005) A block cipher based on a suitable use of the chaotic standard map. Chaos, Solitons Fractals 26(1):117–29
12. Wong KW, Kwok BSH, Law WS (2008) A fast image encryption scheme based on chaotic standard map. Phys Lett 372(15):2645–52
13. Patidar V, Pareek N, Purohit G, Sud K (2011) A robust and secure chaotic standard map based pseudo-random permutation-substitution scheme for image encryption. Opt Commun 284(19):4331–9
14. François M, Grosges T, Barchiesi D, Erra R (2012) A new image encryption scheme based on a chaotic function. Signal Processing: Image Commun 27(3):249–59
15. Zhou Y, Bao L, Chen CP (2013) Image encryption using a new parametric switching chaotic system. Signal Process 93(11):3039–52
16. Pak C, An K, Jang P, Kim J, Kim S (2019) A novel bit-level color image encryption using improved 1D chaotic map. Multimed Tools Appl 78(9):12027–42
17. Xiang H, Liu L (2020) An improved digital logistic map and its application in image encryption. Multimed Tools Appl 79(41):30329–55
18. Zhu B, Liu S, Ran Q (2000) Optical image encryption based on multifractional Fourier transforms. Opt Lett 25(16):1159–61

19. Liu Z, Xu L, Liu T, Chen H, Li P, Lin C et al (2011) Color image encryption by using Arnold transform and color-blend operation in discrete cosine transform domains. Opt Commun 284(1):123–8
20. Kong D, Shen X, Xu Q, Xin W, Guo H (2013) Multiple-image encryption scheme based on cascaded fractional Fourier transform. Appl Opt 52(12):2619–25
21. Chen L, Zhao D, Ge F (2013) Image encryption based on singular value decomposition and Arnold transform in fractional domain. Opt Commun 291:98–103
22. Nayak DR, Dash R, Majhi B (2016) Brain MR image classification using two-dimensional discrete wavelet transform and AdaBoost with random forests. Neurocomputing 177:188–97
23. Dang PP, Chau PM (2000) Image encryption for secure internet multimedia applications. IEEE Trans Consum Electron 46(3):395–403
24. Taneja N, Raman B, Gupta I (2011) Selective image encryption in fractional wavelet domain. AEU-Int J Electron Commun 65(4):338–44
25. Kong D, Shen X (2014) Multiple-image encryption based on optical wavelet transform and multichannel fractional Fourier transform. Opt Laser Technol 57:343–9
26. Luo Y, Du M, Liu J (2015) A symmetrical image encryption scheme in wavelet and time domain. Commun Nonlinear Sci Numer Simul 20(2):447–60
27. Lian S (2009) Efficient image or video encryption based on spatiotemporal chaos system. Chaos, Solitons Fractals 40(5):2509–19
28. Xin G, Fen-lin L, Bin L, Wei W, Juan C (2010) An image encryption algorithm based on spatiotemporal chaos in DCT domain. In: 2010 2nd IEEE International Conference on Information Management and Engineering. IEEE pp 267–70
29. Yu Z, Zhe Z, Haibing Y, Wenjie P, Yunpeng Z (2010) A chaos-based image encryption algorithm using wavelet transform. In: 2010 2nd International conference on advanced computer control. IEEE, vol 2, pp 217–22
30. Ramahrishnan S, Geetha BER et al (2014) Image encryption using chaotic maps in hybrid domain. Int J Commun Comput Technol 2(2):74–8
31. Wang W, Tan H, Pang Y, Li Z, Ran P, Wu J (2016) A novel encryption algorithm based on DWT and multichaos mapping. J Sensors 2016
32. Abdmouleh MK, Khalfallah A, Bouhlel MS (2017) A novel selective encryption DWT-based algorithm for medical images. In: 2017 14th International conference on computer graphics, imaging and visualization. IEEE pp 79–84
33. Zhu S, Zhu C (2019) A new image compression-encryption scheme based on compressive sensing and cyclic shift. Multimed Tools Appl 78(15):20855–5
34. Lu Q, Yu L, Zhu C (2021) A new conservative hyperchaotic system-based image symmetric encryption scheme with DNA coding. Symmetry 13(12):2317
35. Kaur M, Singh D, Kumar V (2022) Improved seven-dimensional (i7D) hyperchaotic map-based image encryption technique. Soft Comput 26(6):2689–2698
36. Wu Y, Noonan JP, Agaian S (2011) NPCR and UACI randomness tests for image encryption. Cyber journals: multidisciplinary journals in science and technology, J Sel Areas Telecommun (JSAT) 1(2):31–38
37. Zhang Y, Xiao D, Wen W, Li M (2014) Breaking an image encryption algorithm based on hyper-chaotic system with only one round diffusion process. Nonlinear Dyn 76(3):1645–1650
38. Brindha M, Gounden NA (2016) A chaos based image encryption and lossless compression algorithm using hash table and Chinese Remainder Theorem. Appl Soft Comput 40:379–390
39. Wang B, Zheng X, Zhou S, Zhou C, Wei X, Zhang Q, Che C (2014) Encrypting thecompressed image by chaotic map and arithmetic coding. Optik 125:6117–6122
40. Alawida M (2023) A novel chaos-based permutation for image encryption. Journal of King Saud University-Computer and Information Sciences 35(6):101595
41. Norouzi B, Seyedzadeh SM, Mirzakuchaki S, Mosavi MR (2014) A novel image encryption based on hash function with only two-round diffusion process. Multimedia Syst 20(1):45–64
42. Zarebnia M, Parvaz R (2020) Dynamical 2D and 3D image encryption method by hybrid system based on cat map and wavelet transform. Optik 219:165148
43. Luo Y, Zhou R, Liu J, Cao Y, Ding X (2018) A parallel image encryption algorithm based on the piecewise linear chaotic map and hyper-chaotic map. Nonlinear Dyn 93:1165–1181
44. Hosny KM, Kamal ST, Darwish MM, Papakostas GA (2021) New image encryption algorithm using hyperchaotic system and fibonacci Q-matrix. Electronics 10(9):1066
45. Hua Z, Zhou Y, Huang H (2019) Cosine-transform-based chaotic system for image encryption. Inf Sci 480:403–419

46. Zhu L, Jiang D, Ni J, Wang X, Rong X, Ahmad M, Chen Y (2022) A stable meaningful image encryption scheme using the newly-designed 2D discrete fractional-order chaotic map and Bayesian compressive sensing. Signal Process 195:108489
47. Dong C (2014) Color image encryption using one-time keys and coupled chaotic systems. Sig Process Image Commun 29:628–640
48. Basu S (2011) International Data Encryption Algorithm (IDEA)-A typical illustration. Journal of Global Research in Computer Science 2:116–118
49. Bansal R, Gupta S, Sharma G (2016) An innovative image encryption scheme based on chaotic map and Vigene‘ re scheme. Multimed Tools Appl
50. Murillo-Escobar M, Cruz-Hernández C, Abundiz-Pérez F, López-Gutiérrez R, Del Campo OA (2015) A rgb image encryption algorithm based on total plain image characteristics and chaos. Signal Process 109:119–131
51. Barker WC, Barker E (2012) Recommendation for the Triple Data Encryption Algorithm (TDEA) block cipher. National Institute of Standards and Technology, Special Publication, pp 800–867
52. Mousa A, Hamad A (2006) Evaluation of the RC4 algorithm for data encryption. Int J Comput Sci Appl 3:44–56

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.